

Sheikh Ahsan Shafique

Full Stack MERN Developer | AI Integration | Final Year Software Engineering Student

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PROFESSIONAL SUMMARY

Passionate Software Engineering student at Bahria University Islamabad with hands-on experience building full-stack MERN applications. Skilled in REST API development, JWT authentication, MongoDB database design, and React frontend development with a strong focus on writing clean, scalable code. Experienced in integrating AI capabilities into web applications including vector embeddings and LLM APIs. Seeking a MERN Stack Internship to contribute to real-world projects and grow as a developer.

EDUCATION

Bahria University, Islamabad | Jan 2023 – Jan 2027

Bachelor of Science in Software Engineering

- Relevant Coursework: Data Structures & Algorithms, Database Management, Operating Systems, Software Construction, Computer Networks, HCI, Artificial Intelligence, Cloud Computing, Software Quality Engineering

TECHNICAL SKILLS

Frontend: React.js, HTML5, CSS3, Bootstrap, Vanilla JavaScript, Responsive UI Design

Backend: Node.js, Express.js, RESTful APIs, JWT Authentication & Authorization, MVC Architecture

Databases: MongoDB Atlas, MySQL, Supabase (PostgreSQL + pgvector)

AI Integration: Google Gemini API, Vector Embeddings, Supabase pgvector, RAG Pipeline, Semantic Search, LLM API Integration

Tools: Git & GitHub, Postman, VS Code, Figma, Vercel, Render, Multer, Cheerio

Languages: JavaScript, Python (basics), Java, C++

Concepts: OOP, Data Structures & Algorithms, Agile Development, Software Design Principles, API Design

PROJECTS

Second Brain – AI-Powered Knowledge Base | *React.js, Node.js, Express.js, MongoDB, Supabase*

pgvector, Google Gemini API

In Development

- Building a full-stack MERN application where users save PDFs, URLs, and notes — backend extracts clean text and stores it in MongoDB with complete JWT auth system.
- Implemented multi-source content ingestion pipeline using Multer for file uploads, Cheerio for URL scraping, and pdf-parse for PDF text extraction.
- Integrated Google Gemini Embedding API and Supabase pgvector — content is chunked using LangChain text splitter and stored as vectors for semantic similarity search.
- Built full RAG chat pipeline — user question converted to vector, similar chunks retrieved from Supabase, injected as context into Gemini Flash, AI answers with source citations.
- Designed RESTful API with protected routes, authorization checks, and proper error handling across all endpoints.

Relaxa – AI-Based Mental Wellness Web Application | *React.js, Node.js, Express.js, MongoDB*

- Developed a full-stack MERN mental wellness platform with mood tracking, wellness exercises, AI chatbot, and mood analysis reports.

- Applied AI search algorithms (BFS, A*) for intelligent stress-relief recommendations and state-space modeling.
- Designed responsive frontend and scalable backend with modern UI/UX principles.

Live Demo: <https://relaxa-one.vercel.app>

CivicConnect – Smart City Resident Services Portal | *React.js, Node.js, Express.js, MongoDB*

- Built during AUREX AI 2026 Web Development Competition at Bahria University — full-stack civic portal with complaints, permits, and service request tracking.
- Implemented role-based dashboards for Residents, Department Staff, Admins, and Super Admins with secure authentication and request lifecycle tracking.
- Integrated analytics dashboards, SLA monitoring, reusable components, and scalable backend APIs.

Campus Notice Board – University Announcements Platform | *React.js, Supabase, PostgreSQL*

- Full-stack notice board with real-time updates, category filtering, and role-based access using Supabase real-time subscriptions.
- Deployed on Vercel with Supabase cloud handling all backend infrastructure including authentication and database.

Live Demo: <https://campusnotice-board.vercel.app>

E-Commerce Toy Store Platform | *Node.js, Express.js, MySQL, HTML5, CSS3, JavaScript*

- Full-stack e-commerce platform with checkout workflows, order tracking, review system, and admin dashboard.
- Designed RESTful APIs, relational database schema, and environment management for scalable backend operations.

ACHIEVEMENTS & EXPERIENCE

- Currently building a production-grade RAG system with real vector embeddings and LLM API integration on top of MERN stack.
- Participated in AUREX AI 2026 Web Development Competition organized by MLSA BSEAS at Bahria University.
- Built and deployed multiple full-stack MERN applications with live URLs.
- Consistently practicing Data Structures & Algorithms and software engineering best practices.